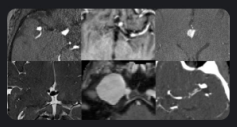
RADIOLOGICAL SOCIETY OF NORTH AMERICA · FEATURED CODE COMPETITION · 2 MONTHS TO GO

Submit Prediction

RSNA Intracranial Aneurysm Detection

Detect the presence and location of intracranial aneurysms in multimodal imaging data



Dataset Description

The competition data comprises several thousand expertly-curated medical imaging series of the brain from a variety of modalities, including CTA, MRA, and T1 post-contrast and T2-weighted MRI. The series have been labeled in thirteen locations for the presence of aneurysms. We include spatial localization labels for all aneurysms and vessel segmentations for a subset of the cases.

File and Field Information

train.csv This file contains the primary training labels. Each row corresponds to a single image series. The other columns provide more granular labels for the location of any aneurysms as well as basic demographic data.

- `SeriesInstanceUID` - A unique identifier for each scan series.
- `Modality` - Identifies the mode of imaging: CTA, MRA, MRI, etc.
- `PatientAge` and `PatientSex` - Demographic data for the subject of the scan.
- `Left Infraclinoid Internal Carotid Artery` - One of thirteen binary variables indicating presence of an aneurysm in this location. The other twelve follow. In all cases, `1` denotes the presence of an aneurysm and `0` denotes its absence.
- `Right Infraclinoid Internal Carotid Artery`
- `Left Supraclinoid Internal Carotid Artery`
- `Right Supraclinoid Internal Carotid Artery`
- `Left Middle Cerebral Artery`
- `Right Middle Cerebral Artery`
- `Anterior Communicating Artery`
- `Anterior Communicating Artery`
- `Left Anterior Cerebral Artery`
- `Right Anterior Cerebral Artery`
- `Left Posterior Communicating Artery`
- `Right Posterior Communicating Artery`
- `Basilar Tip`
- `Other Posterior Circulation`
- `Aneurysm Present` The main target variable, binary, indicating if an aneurysm is present anywhere in the scan.

train_localizers.csv This file provides localization data for the aneurysms in the training set. For each aneurysm, it provides the specific image (`SOPInstanceUID`) and the coordinates near the aneurysm's center.

- `SeriesInstanceUID` - A unique identifier for each scan series, corresponding to that in `train.csv`.
- `SOPInstanceUID` - A unique identifier for a specific image within a series.
- `coordinates` - The xy-coordinates of the center of the aneurysm in the image.
- `location` - A text description of the aneurysm's location.

series/ Contains training-set DICOM series, with one series per folder. The filepaths have the form `series/{SeriesInstanceUID}/{SOPInstanceUID}.dcm`.

The DICOM files contain the identifier tags: `PatientID`, `StudyUID`, and `SeriesUID`. All three tags are included in the DICOMs for conformity to the DICOM standard, but only `SeriesUID` is meaningful. The `PatientID` and `StudyUID` tags do not identify anything.

segmentations/ Contains NifTI vessel segmentations for a subset of the DICOM series in `series/`. The labels are described as follows:

Label Value	Label
1	Other Posterior Circulation
2	Basilar Tip
3	Right Posterior Communicating Artery
4	Left Posterior Communicating Artery
5	Right Infraclinoid Internal Carotid Artery
6	Left Infraclinoid Internal Carotid Artery
7	Right Supraclinoid Internal Carotid Artery
8	Left Supraclinoid Internal Carotid Artery
9	Right Middle Cerebral Artery
10	Left Middle Cerebral Artery
11	Right Anterior Cerebral Artery
12	Left Anterior Cerebral Artery
13	Anterior Communicating Artery

kaggle_evaluation/ Files used by the evaluation API. See the demo submission for an illustration of how to use the API. This API will serve the test set to your submission one series at a time, in unfixed random order. There are nearly 2500 series in the test set.

The DICOM tags in the test set have been restricted to: `['BitsAllocated', 'BitsStored', 'Columns', 'FrameOfReferenceUID', 'HighBit', 'ImageOrientationPatient', 'ImagePositionPatient', 'InstanceNumber', 'Modality', 'PatientID', 'PhotometricInterpretation', 'PixelRepresentation', 'PixelSpacing', 'PlanarConfiguration', 'RescaleIntercept', 'RescaleSlope', 'RescaleType', 'Rows', 'SOPClassUID', 'SOPInstanceUID', 'SamplesPerPixel', 'SliceThickness'`

Files
1013717 files

Size
314.53 GB

Type
dcm, nii, py + 2 others

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'SpacingBetweenSlices', 'StudyInstanceUID', 'TransferSyntaxUID'].

train.csv (474.21 kB)

Detail Compact Column

10 of 18 columns

SeriesInstanceUID	PatientAge	PatientSex	Modality	Left Infraclinoid I...	Right I
4405 unique values	18 – 89	Female 69% Male 31%	CTA 42% MRA 29% Other (1292) 29%	0 – 1	0
1.2.826.0.1.3680043.8.498.10004044420235051083751528781076556647	64	Female	MRA	0	0
1.2.826.0.1.3680043.8.498.10004684224894397679901841656954650085	76	Female	MRA	0	0
1.2.826.0.1.3680043.8.498.10005158603912009425635473100344077317	58	Male	CTA	0	0
1.2.826.0.1.3680043.8.498.10009383100068795488741533244914370182	71	Male	MRA	0	0
1.2.826.0.1.3680043.8.498.10012790035410518400400834395242853657	48	Female	MRA	0	0
1.2.826.0.1.3680043.8.498.10014757658335054766479957992112625961	53	Female	CTA	0	0
1.2.826.0.1.3680043.8.498.10021411248005513321236647460239137906	55	Female	CTA	0	0

Data Explorer

314.53 GB

- kaggle_evaluation
- segmentations
- series
- train.csv
- train_localizers.csv

Summary

- 1.01m files
- 23 columns

Download All

DOWNLOAD DATA

```
>_ kaggle competitions download -c rsna-intracranial-aneurysm-detection
```

Metadata

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